THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of y in $13x + 5y = 67$, $6x + 7y = 45$ is: 1

(a) 3 **(b)** -3 **(c)** 4 **(d)** 7

2. The product of the roots of $3x^2 + 7x - 6 = 0$ is: 1

(a) -2 **(b)** $\frac{7}{3}$ **(c)** 2 **(d)** $-\frac{7}{3}$

3. The largest number that divides 74 and 130 leaving remainders 2 and 4 respectively is: 1

(a) 72 **(b)** 9 **(c)** 18 **(d)** 36

4. If $x = 3$ is a zero of $x^2 - 8x + k$, then k is: 1

(a) 15 **(b)** 33 **(c)** 5 **(d)** 24

5. The pair $3x + ky = 9$, $6x + 6y = 18$ has infinitely many solutions when $k =$ 1

(a) 9 **(b)** 4 **(c)** 3 **(d)** 6

6. The roots of $x^2 - 2x - 15 = 0$ differ by: 1

(a) 15 **(b)** 9 **(c)** 2 **(d)** 8

7. The least number which when divided by 4, 6 and 10 leaves remainder 2 each time is: 1

(a) 60 **(b)** 122 **(c)** 62 **(d)** 58

8. The value of k for which $x = 3$ is a zero of $x^2 + kx - 6$ is: 1

(a) 5 **(b)** 1 **(c)** -1 **(d)** -3

9. For the pair $kx + 3y = 4$ and $8x + 2y = 5$ to have a unique solution, k must NOT equal: 1

(a) 12 **(b)** 11 **(c)** $\frac{32}{5}$ **(d)** $\frac{16}{3}$

10. If $x = 5$ is a root of $x^2 - 11x + k = 0$, then $k =$ 1

(a) -30 **(b)** 80 **(c)** 6 **(d)** 30

11. The decimal expansion of $\frac{129}{312500}$ is: 1

(a) terminating **(b)** an integer

(c) non-terminating repeating **(d)** non-terminating non-repeating

12. If the zeroes of $x^2 + px + q$ are 2 and 7, then p equals: 1

(a) -14 **(b)** -9 **(c)** 14 **(d)** 9

13. If $4x + 3y = 31$ and $3x + 4y = 25$, then $x + y =$ 1

(a) 6 **(b)** 28 **(c)** 8 **(d)** 56

14. The equation $x^2 + 4x + 4 = 0$ has: 1

(a) two distinct real roots **(b)** no real roots
(c) two equal real roots **(d)** three real roots

15. If $\sqrt{5}$ is irrational, then $\sqrt{5} \times \sqrt{5}$ is: 1

(a) non-terminating non-repeating **(b)** not defined
(c) irrational **(d)** rational

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find k so that the pair $2x + ky = 6$ and $4x + 4y = 12$ has infinitely many solutions. 2

17. Find k so that $kx^2 - 6x + 1 = 0$ has equal roots. 2

18. Find the largest number which divides 398 and 436 leaving remainders 7 and 11 respectively. 2

19. If α, β are zeroes of $x^2 - 10x + 21$, find $\alpha^2 + \beta^2$. 2

20. Solve by the cross-multiplication method: $5x + 3y = 29$ and $4x + 7y = 37$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Solve by elimination: $5x + 3y = 29$ and $4x + 7y = 37$. 3

22. Solve for x: $\dfrac{1}{x} + \dfrac{1}{x+1} = \dfrac{7}{12}$. 3

23. Three bells toll at intervals of 6, 10 and 14 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

24. Find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $x^2 - 7x + 6$. 3

OR

The sum of the digits of a two-digit number is 13. If the digits are interchanged, the number becomes 45 more. Find the number. 3

25. Find the nature of the roots of $2x^2 + 5x + 4 = 0$ and, if real, find them. 3

OR

Show that $(7 + 3\sqrt{2})(7 - 3\sqrt{2})$ is rational but $7 + 3\sqrt{2}$ is irrational. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two tankers contain 612 litres and 918 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

Find all the zeroes of $x^3 - 6x^2 + 3x + 10$ if two of its zeroes are -1 and 2. 5

27. A father is 6 times as old as his son. After 10 years the father will be 5 times as old as the son will be. Find their present ages. 5

OR

A train travels 280 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the journey. Find the original speed. 5

END OF QUESTION PAPER — **Total: 50 Marks**